Executive Business Report

Sales Forecasting and Demand Analysis for Superstore Sales

Executive Summary

This project analyzed the company's historical sales data to understand sales trends, predict future sales, identify unusual sales patterns, and group products based on their demand. Three forecasting models (SARIMA, Prophet, and XGBoost) were tested, and XGBoost gave the most accurate predictions. The project also identified sales anomalies and grouped products into demand segments to support better inventory planning. An interactive dashboard was created to help users explore sales performance, forecasts, and product demand in an easy and visual way.

Key Findings from Data Analysis and Forecasting

- Sales showed an overall increasing trend over the four years, although monthly sales changed because of seasonal demand.
- Technology and Office Supplies contributed a large share of total sales.
- The West and East regions generated higher sales than the other regions.
- XGBoost was the best forecasting model because it had the lowest prediction errors (MAE, RMSE, and MAPE) compared to SARIMA and Prophet.
- Based on the forecast, sales are expected to remain stable over the next three months with normal month-to-month changes.

Three-Month Sales Forecast

The forecasting model predicts that sales will remain healthy over the next three months. While small increases and decreases are expected, there is no sign of a major drop in sales.

The confidence range means the actual sales may be slightly higher or lower than the predicted value, but they are expected to stay close to the forecast under normal business conditions.

Top 3 Anomalies Detected

1. Sudden Sales Spike

A very high sales week was detected.

Possible reason: A festive season, holiday sale, promotional campaign, or bulk customer orders.

2. Unusually Low Sales

A week with much lower sales than normal was identified.

Possible reason: Low customer demand, stock shortages, or fewer business days due to holidays.

3. Consecutive High Sales Weeks

Several weeks showed sales much higher than the normal pattern.

Possible reason: Seasonal shopping periods, successful discounts, or special marketing campaigns.

Product Demand Segmentation

Products were grouped into different demand categories using K-Means clustering.

- **High Volume, Stable Demand**

These products sell regularly and consistently.

Recommendation: Keep higher inventory levels to avoid stock shortages.

- **Growing Demand**

Demand for these products is increasing over time.

Recommendation: Increase stock gradually and monitor future demand closely.

- **Low Volume, High Volatility**

Sales change frequently and are difficult to predict.

Recommendation: Maintain limited inventory and restock only when required.

- **Declining Demand**

Demand for these products is decreasing.

Recommendation: Reduce inventory levels and avoid overstocking.

Business Recommendations

1. Use XGBoost for Sales Forecasting

XGBoost produced the lowest forecasting errors among all three models, making it the most reliable option for predicting future sales.

2. Plan Inventory Based on Demand Segments

Products with stable and growing demand should be stocked more, while declining or highly unpredictable products should have lower inventory levels.

3. Monitor Sales Anomalies Regularly

Large changes in sales should be reviewed quickly to understand whether they are caused by promotions, seasonal demand, or operational issues. This can improve future planning and decision-making.

Risk / Limitation

The forecasting models are trained using historical sales data. If unexpected events occur, such as economic changes, supply chain disruptions, new competitors, or sudden changes in customer behavior, the predictions may become less accurate. The model should be updated regularly with new sales data to maintain its performance.

Conclusion

This project shows how machine learning and data analysis can help businesses understand sales trends, predict future demand, identify unusual sales behavior, and improve inventory management. The XGBoost model provides the best forecasting performance and is recommended for business use. The interactive Streamlit dashboard makes it easy for managers to monitor sales and make informed decisions based on data.